

Determining which side

1 Which of these triangles are right-angled triangles? Tick 3 correct answers

a

b

c

d

e

☐ a

☒ b

☒ c

☐ d

☒ e

2 In the diagram, $x = 101$, $y = 45$, $z = 56$. Which of these angles is the right angle? Tick 1 correct answer

☐ a°

☒ b°

☐ c°

☐ impossible to tell, but this is a right-angled triangle

☐ this is not a right-angled triangle

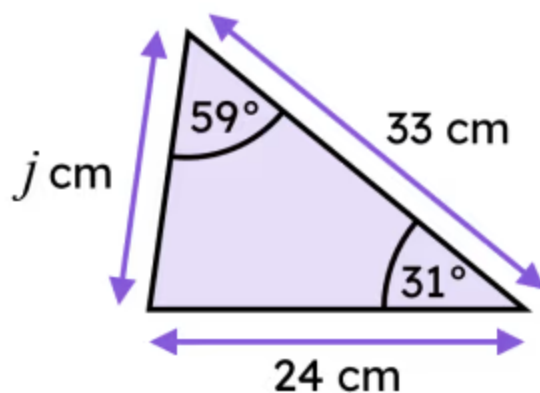
- 3 A right-angled triangle has sides of length 21 inches, 220 inches, and 221 inches. Which of these are correct Pythagoras' theorem equations for this triangle? Tick **2** correct answers

- ☐ $21 + 220 = 221$
- ☒ $21^2 + 220^2 = 221^2$
- ☒ $221^2 = 220^2 + 21^2$
- ☐ $220^2 = 221^2 + 21^2$
- ☐ $221^2 + 21^2 = 220^2$

- 4 A right-angled triangle has sides of length 39 inches, 80 inches, and 89 inches. Which of these are correctly rearranged Pythagoras' theorem equations for this triangle? Tick **3** correct answers

- ☒ $89^2 - 39^2 = 80^2$
- ☐ $89^2 + 39^2 = 80^2$
- ☒ $89^2 - 80^2 = 39^2$
- ☐ $89^2 = 80^2 - 39^2$
- ☒ $0 = 89^2 - 80^2 - 39^2$

- 5 Using Pythagoras' theorem, calculate the length of the side labelled j , rounded to 1 decimal place. Length of side j is 22.6 cm. Fill in the blank



- 6 Using Pythagoras' theorem twice, calculate the length of the side labelled h , rounded to 2 decimal places. Length of side h is 20.69 cm. Fill in the blank

